

18.2.3 Differentiable Convex and Concave Functions

Now, we focus on differentiable convex and concave functions. Let $f : S \rightarrow R$ defined on an open convex set S of R^n is said to be differentiable if the gradient

$$\nabla f(u) = \left(\frac{\partial f(u)}{\partial u_1}, \frac{\partial f(u)}{\partial u_2}, \frac{\partial f(u)}{\partial u_3}, \dots, \frac{\partial f(u)}{\partial u_n} \right) \text{ exists at each } u \in S.$$

A) Second Order Derivative Test for Convexity and Concavity: The Two Variables Case

A function $A = f(u, v)$ is defined over the open set P in the plane with continuous partial derivatives of first and second order. Then,

$$\text{a) } f \text{ is concave} \Leftrightarrow f''_{uu} \leq 0, f''_{vv} \leq 0 \text{ and } \begin{vmatrix} f''_{uu} & f''_{uv} \\ f''_{vu} & f''_{vv} \end{vmatrix} \geq 0$$

$$\text{b) } f \text{ is convex} \Leftrightarrow f''_{uu} \geq 0, f''_{vv} \geq 0 \text{ and } \begin{vmatrix} f''_{uu} & f''_{uv} \\ f''_{vu} & f''_{vv} \end{vmatrix} \geq 0$$

where all inequalities should hold throughout A

For testing **strictly concave and strictly convex**, let $A = f(x, y)$ be a function defined over an open set P in the plane with continuous partial derivatives of first and second order. Then,

$$\text{a) } f \text{ is strictly concave} \Leftrightarrow f''_{uu} < 0 \text{ and } \begin{vmatrix} f''_{uu} & f''_{uv} \\ f''_{vu} & f''_{vv} \end{vmatrix} \geq 0$$

$$\text{b) } f \text{ is strictly convex} \Leftrightarrow f''_{uu} > 0 \text{ and } \begin{vmatrix} f''_{uu} & f''_{uv} \\ f''_{vu} & f''_{vv} \end{vmatrix} \geq 0,$$

where all in equalities should hold throughout A .

B) Second Order Derivative Test for Convexity and Concavity: The n Variable Case

Let S be a nonempty set in R^n , and let $f : S \rightarrow R$. Then, f is said to be twice differentiable if there exist a vector function $\nabla f(u)$ and a $n \times n$ symmetric matrix $H(u)$, called the Hessian matrix.

For twice differentiable function the Hessian matrix $H(u)$ is given comprised of the 2nd order partial derivatives and is denoted as

$$H(u) = \begin{bmatrix} \frac{\partial^2 f(u)}{\partial u_1^2} & \dots & \frac{\partial^2 f(u)}{\partial u_1 \partial u_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(u)}{\partial u_n \partial u_1} & \dots & \frac{\partial^2 f(u)}{\partial u_n^2} \end{bmatrix} = \begin{bmatrix} f_{11}(u) & f_{12}(u) & \dots & f_{1n}(u) \\ f_{21}(u) & f_{22}(u) & \dots & f_{2n}(u) \\ \dots & \dots & \dots & \dots \\ f_{n1}(u) & f_{n2}(u) & \dots & f_{nn}(u) \end{bmatrix}$$

Then determinants are called the leading principal minors of $H(u)$ denoted by $D_k(u)$.

f is concave $\Leftrightarrow$ for all $D_k(u), (-1)^k D_k(u) \geq 0$ for $k = 1, \dots, n$.

f is convex $\Leftrightarrow$ for all $D_k(u), D_k(u) \geq 0$ for $k = 1, \dots, n$.

Theorem

Suppose $S \neq \emptyset$ be set in R^n , and let $f : S \rightarrow R$ be two times differentiable on S then, f is said to be convex on S if the Hessian matrix is positive semi definite (PSD) at each point in S ; that is, for any $\bar{u}$ in S , we have $u' H(\bar{u}) u \geq 0$ for all $u \in R^n$. Symmetrically, a function f is concave on S if its Hessian matrix is negative semi definite (NSD) everywhere in S , i.e., for any $\bar{u} \in S$, we have $u' H(\bar{u}) u \leq 0$ for all $u \in R^n$.

18.2.4 Quasi-Convex and Quasi-Concave Functions

Generalizations of a Convex Function

In this section, we will show various types of functions which are like convex and concave functions but that share only some of their desirable properties. We will learn that many of the results do not require the restrictive assumptions of convexity, but rather the less restrictive assumptions of quasi-convexity, quasi concave.

Let f be given function defined on a convex subset S of a real vector space is quasi-convex if for any $u, v \in S$ and $\theta \in [0, 1]$

$$f(\theta u + (1 - \theta)v) \leq \max \{f(u), f(v)\}$$

Further more if

$$f(\theta u + (1 - \theta)v) < \max \{f(u), f(v)\}$$

For any $u \neq v$ and $\theta \in [0, 1]$, then f is strictly quasi-convex.

The function f is said to be quasi concave if $-f$ is quasi-convex and a strictly quasi-concave functions a function whose negative is strictly quasi-convex. Equivalently a function f is quasi-concave if

$$f(\theta u + (1 - \theta)v) \geq \min \{f(u), f(v)\}$$

and strictly quasi-concave if

$$f(\theta u + (1 - \theta)v) > \min \{f(u), f(v)\}$$

Examples of Quasi-Convex Functions:

- $\sqrt{|u|}$ is quasi - convex on R .
- $\log u$ is quasi-convex on R_+

A) Differentiable Quasi-Convex and Quasi-Concave Functions

Suppose S be a nonempty open convex set in R^n , and function $f: S \rightarrow R$ be differentiable on S . Then, f is quasi convex if either one of the following equivalent statements holds:

- If $u, v \in S$ and $f(u) \leq f(v)$, then $\nabla f(v)^T(u-v) \leq 0$.
- If $u, v \in S$ and $\nabla f(v)^T(u-v) > 0$, then $f(u) > f(v)$.

To illustrate the above theorem, let $f(u) = u^3$.

To check quasi-convexity, suppose $f(u_1) \leq f(u_2) \Rightarrow u_1^3 \leq u_2^3$. This is true only if $u_1 \leq u_2$.

Now, consider $\nabla f(u_2)(u_1 - u_2) = 3(u_1 - u_2)u_2^2$. Since $u_1 \leq u_2$, $3(u_1 - u_2)u_2^2 \leq 0$

Therefore, $f(u_1) \leq f(u_2) \Rightarrow \nabla f(u_2)(u_1 - u_2) \leq 0$ and hence by the above theorem, f is quasi-convex.

Consider $f(u_1, u_2) = u_1^3 + u_2^3$. Let $u_1 = (2, -2)^T$ and $u_2 = (1, 0)^T$. $f(u_1) = 0$ and $f(u_2) = 1$, so that $f(u_1) < f(u_2)$.

But $\nabla f(u_2)^T(u_1 - u_2) = (3, 0)(1, -2)^T = 3$. By the necessary part of the theorem, f is not quasi convex. This shows that the sum of two quasi-convex functions is not necessarily quasi-convex.

Propositions:

- A function which is both quasi-convex and quasi-concave is called quasi-linear.
- Every convex function is quasi-convex.
- A concave function can be quasi-convex function. For example, $\log(u)$ is concave, and it is quasi-convex.
- If $f(u)$ and $g(v)$ are positive convex decreasing functions, then, $f(u)g(v)$ is quasi-convex.
- Sum of two Quasi-convex functions is not necessarily a Quasi-convex.

B) Strongly Quasi-Convex and Quasi-Concave Functions

The concept of strong convexity extends the notion of strict convexity. Let S be a nonempty open convex set in R^n , and let $f: S \rightarrow R$. The function f is said to be strongly quasi convex if for each we have $u, v \in S$, we have

$$f(\theta u + (1-\theta)v) < \max\{f(u), f(v)\} \text{ for each } \theta \in (0,1), \text{ with } u \neq v.$$

The function f is said to be strongly quasi-concave if $-f$ is strongly quasi-convex. From the definition of convex function, quasi-convex function and strictly quasi-convex function, the following statements hold:

- Every strictly convex function is strongly quasi-convex.

- Every strongly quasi-convex function is strictly quasi-convex.
- Every strongly quasi-convex function is quasi-convex

Check Your Progress 1

- 1) Find out the convexity and concavity of the following functions over the set of real numbers that are non- negative

a. $f(u) = 2u_1^3 - 6u_2^2$ b. $f(u) = -8u^2$

.....

- 2) Do you think the idea of quasi concavity is weaker than the idea of concavity?

.....

18.3 CONSTRAINED OPTIMISATION – NON-DEGENERATE CONSTRAINT QUALIFICATIONS (NDCQ)

Optimisation of functions of multiple variables subjected to equality constraints using the method of constrained variation, the method of Lagrange multipliers.

Minimize (or maximize) $f(u)$

Subject to $g_i(u) \leq 0 \quad i = 1, 2, \dots, m$...(18.2)

and $h_j(u) = 0 \quad j = 1, 2, \dots, l$

where $u = (u_1, u_2, \dots, u_n)^T$ is a column vector of n real-valued design variables. In equation (18.2), $f: R^n \rightarrow R$ is the objective or cost function $g_i: R^n \rightarrow R$ are inequality constraints and $h_j: R^n \rightarrow R$ are equality constraints. A vector $x \in R^n$ satisfying all the constraints is called a *feasible solution* to the problem. The collection of all such solutions forms the *feasible region*. The nonlinear programming problem, then, is to find a feasible point x' such that $f(x) \geq f(x')$ for each feasible point x . Such a point x' is called an *optimal solution*. We define the feasible set Ω as

$$\Omega = \{x \in R^n : h_j = 0, g_i(x) \leq 0\}$$

The phrase ‘minimization problem’ refers to finding a local minimizer in the feasible set Ω . The nonlinear programming problems having more than two variables cannot be solved using graphical method. In such cases we have to use the method of Lagrange multipliers.

The above analysis helps us in obtaining the following generalization of the n -dimensional minimization problem with equality constraints:

$$\max_{u_1 \in R} f(u_1, \dots, u_n)$$

Subject to, $g_1(u_1, u_2, \dots, u_n) \leq b_1, \dots, g_k(u_1, u_2, \dots, u_n) \leq b_k,$

$$h_1(u_1, u_2, \dots, u_n) = c_1, \dots, h_m(u_1, u_2, \dots, u_n) = c_m.$$

The function $f(u)$ is called the objective function, $g(u)$ is called an inequality constraint, and $h(u)$ is called an equality constraint. In the above problem there are k inequality constraints and m equality constraints. In the following we will always assume that f , g and h are C^1 functions, i.e. they are differentiable and their derivatives are continuous. Notice that this problem differs from the regular unconstrained optimisation problem in that instead of finding the maximum of $f(u)$, we are finding the maximum of $f(u)$ only over the points which satisfy the constraints.

Example 18.1: Maximize $f(u) = u^2$ subject to $0 \leq u \leq 1$.

Ans. We know that $f(u)$ is strictly monotonically increasing over the domain, therefore the maximum (if it exists) must lie at the largest number in the domain. Since we are optimizing over a compact set, the point $u=1$ is the maximal number in the domain, and therefore it is the maximum.

A. Case of One Constraint

Consider a simple optimisation problem with only one constraint:

$$\max_{u_1 \in R} f(u_1, \dots, u_n)$$

Subject to

$$h(u_1, u_2, \dots, u_n) = c.$$

Now draw level sets of the function $f(u_1, u_2, \dots, u_n)$. Since we may not be able to achieve the unconstrained maxima of the function due to our constraint, we seek the value of u which gets us onto the highest level curve of $f(u)$ while remaining on the function $h(u)$. Notice also that the function $h(u)$ will be just tangent to the level curve of $f(u)$.

Let us denote the point which maximizes the optimisation problem u^* , (also referred to as the maximizer). Since at u^* the level curve of $f(u)$ is tangent to the curve $g(u)$, it must also be the case that the gradient of $f(u^*)$ must be in the same direction as the gradient of $h(u^*)$, or

$$\nabla f(u^*) = \lambda \nabla h(u^*),$$

where λ is merely a constant.

Now let us define the Lagrangian the following way:

$$L(u_1, u_2, \dots, u_n, \lambda) = f(u_1, \dots, u_n) - \lambda[h(u_1, \dots, u_n) - c]$$

Setting the partial derivatives of the above function with respect to u equal to zero will yield the first order conditions for a constrained maximum:

$$\nabla f(u^*) - \lambda \nabla h(u^*) = 0.$$

Setting the partial derivative with respect to λ equal to zero gives us our original constraint back:

$$h(u^*) - c = 0.$$

So, the first order conditions for this problem are simply $\nabla L(u, \lambda) = 0$

B. Case of Several Constraints

Consider the problem

$$\max_{u_1 \in R} f(u_1, \dots, u_n)$$

Subject to

$$h_1(u_1, u_2, \dots, u_n) = c_1, \dots, h_m(u_1, \dots, u_n) = c_m.$$

Let us first talk about how the Lagrangian approach might fail. As we saw for one constraint, if $\nabla h(u^*) = 0$ then the constraint drops out of the equation. Now consider the Jacobian matrix, or a vector of the gradients of the different $h_i(u^*)$.

$$Dh(u^*) = \begin{pmatrix} \nabla h_1(u^*) \\ \vdots \\ \nabla h_m(u^*) \end{pmatrix} = \begin{pmatrix} \frac{dh_1(u^*)}{du_1} & \cdot & \cdot & \cdot & \frac{dh_1(u^*)}{du_n} \\ \vdots & \cdot & \cdot & \cdot & \cdot \\ \frac{dh_m(u^*)}{du_1} & \cdot & \cdot & \cdot & \frac{dh_m(u^*)}{du_n} \end{pmatrix}$$

Notice that if any of the $\nabla h_i(u^*) = 0$, then that constraint will not be considered in the analysis. Also, there will be a row of zeros in the Jacobian, and therefore the Jacobian will not be of full rank. The generalization of the condition that $\nabla h(u^*) \neq 0$ for the case when $m=1$ is that the Jacobian matrix must be of rank m . Otherwise, one of the constraints is not being considered, and the analysis fails. We call this condition the **non-degenerate constraint qualification (NDCQ)**.

Note that we only must check whether the NDCQ holds at points in the constraint set, since points outside the constraint set are not solution

candidates anyway. If we test for NDCQ and find that the constraint is violated for some point within our constraint set, we must add this point to our candidate solution set. The Lagrangian technique simply does not give us any information about this point.

The Lagrangian for the multi-constraint optimisation problem is

$$L(u_1, \dots, u_n, \lambda) = f(u_1, \dots, u_n) - \sum_{i=1}^m \lambda_i [h_i(u_1, \dots, u_n) - c_i]$$

Therefore, the necessary conditions for a maximum are

$$\frac{dL}{du_1} = 0, \dots, \frac{dL}{du_n} = 0$$

$$\frac{dL}{d\lambda_1} = 0, \dots, \frac{dL}{d\lambda_n} = 0.$$

Check Your Progress 2

1) Does NDCQ hold for the following?

$$\min u_1^2 + u_2^2 + u_3^2$$

$$\text{Subject to } 3u_1 + u_2 + u_3 = 5$$

$$u_1 + u_2 + u_3 = 1.$$

.....
.....
.....

2) Does NDCQ hold for the following?

$$\max 3 + u_2 u_3$$

$$\text{Subject to } u_2^2 + u_3^2 = 1.$$

.....
.....
.....

3) Solve the following:

$$\text{Max } U_1 U_2 U_3$$

$$\text{Subject to } U_1 + U_2 + U_3 = c^2$$

$$\text{Given } U_1 \geq 0, U_2 \geq 0 \text{ and } U_3 \geq 0.$$

.....
.....

18.4 OPTIMISATION WITH INEQUALITY CONSTRAINT(S) - BINDING AND NON- BINDING CONSTRAINTS

A. Inequality Constraint Optimisation for One Constraint

Consider the problem

$$\max_{u_1 \in R} f(u_1, \dots, u_n)$$

Subject to

$$g(u_1, u_2, \dots, u_n) \leq b.$$

In this case the solution is not constrained to the curve $g(u)$, but merely bounded by it.

In order to understand the new conditions, imagine the graph of the level sets which we talked about before. Instead of being constrained to the function $g(u)$, the domain is now bounded by it. However, the boundary of the function is still the same as before. Notice that there is still a point where the boundary is tangent to some level set of $g(u)$. The question now is whether the boundary is binding or not binding.

Remember that we are trying to find candidate points for a global maximum. Restricting our original domain X to the set of points where $g(u) \leq b$ gives us two types of candidate points: Points on the boundary $g(u) = b$ where $g(u) = b$ is tangent to the level curves of $f(u)$, and local maxima of $f(u)$ for which $g(u) \leq b$. The first type we can find by the constrained FOC $\nabla f(u) = \lambda \nabla g(x)$, and the second type we can find by the unconstrained FOC $\nabla f(u) = 0$. Let us look at each of these one by one.

Case 1: Candidates on the boundary (constraint binding)

This is the case where an unconstrained maximum lies outside of the constraint set. In other words, the inequality constrains us from reaching a maximum of f . In this case, the gradient of $f(u)$ is going to point in the steepest direction up the graph. The gradient of $g(x)$ points to the set $g(u) \geq b$ (since it points in the direction of increasing $g(u)$). Therefore $\nabla g(u)$ is pointing in the same direction as $\nabla f(u)$, which implies that $\lambda \geq 0$. So necessary conditions for a solution on the boundary are

$$\nabla f(u) - \lambda \nabla g(x) = 0 \quad \text{and} \quad \lambda \geq 0.$$

Case 2: Candidates not on the boundary (constraint not binding)

This is the case where an unconstrained maximum lies inside the constraint set. In other words, the inequality does not constrain us from reaching this

maximum of f . The first order condition is simply $\nabla f(u) = 0$, which we can rewrite (to take the same shape as above) as

$$\nabla f(u) - \lambda \nabla g(u) = 0 \text{ and } \lambda = 0.$$

In summary, either the constraint is binding (tight), that is $g(u) - b = 0$ and $\lambda \geq 0$, or the constraint is not-binding (slack), and then $\lambda = 0$. We can summarize this new set of conditions as what is called the ‘complementary slackness conditions’

$$[g(u) - b]\lambda = 0$$

This works because if the constraint is binding, then $g(u) - b = 0$, and if the constraint is not binding, then we want to ignore it by having $\lambda = 0$. As we can see, it does not matter whether the constraint binds or does not bind, the Lagrangian multiplier must always be greater than or equal to 0. Therefore, another new set of conditions are

$$\lambda_i \geq 0, \forall i.$$

Finally, we have included our original inequality constraint $g(u) \leq b$.

We can summarize all these FOCs in terms of the Lagrangian, which we define as before to be $L(u, \lambda) = f(u) - \lambda(g(u) - b)$:

$$\frac{\partial L(u, \lambda)}{\partial u_i} = \frac{\partial f(u)}{\partial u_i} - \lambda \frac{\partial g(u)}{\partial u_i} = 0 \text{ for all } i = 1, \dots, n$$

$$\lambda \frac{\partial L(u, \lambda)}{\partial \lambda} = \lambda [g(u) - b] = 0 \text{ complementary slackness}$$

$$\frac{\partial L(u, \lambda)}{\partial \lambda} = [g(u) - b] \leq 0 \text{ original constraint}$$

$$\lambda \geq 0$$

B. Inequality Constraint Optimisation for Several Constraints

Consider the problem:

$$\max_{u_1 \in R} f(u_1, \dots, u_n)$$

Subject to

$$g_1(u_1, u_2, \dots, u_n) \leq b_1, \dots, g_k(u_1, \dots, u_n) \leq b_k$$

In order to understand the new NDCQ, we must realize that if a constraint does not bind, we do not care if it drops out of the equation. The point of the NDCQ was to ensure that binding constraints do not drop out. Therefore, the NDCQ with inequality constraints is the same as the equality constraints, except for the fact that we only care about the Jacobian matrix of the binding constraints, or the Jacobian for the constraints with $\lambda_i > 0$. (Notice that we cannot tell in advance which constraints will be binding, so we need to check

all of them when we check the NDCQ before computing the solution candidates.)

The following first order conditions will tell us the candidate points for a maximum

$$\frac{dL}{du_1} = 0, \dots, \frac{dL}{du_n} = 0$$

$$[g_1(u_1, u_2, \dots, u_n) - b_1]\lambda_1 = 0, \dots, [g_k(u_1, \dots, u_n) - b_k]\lambda_k = 0$$

$$g_1(u_1, u_2, \dots, u_n) \leq b_1, \dots, g_k(u_1, \dots, u_n) \leq b_k$$

$$\lambda_i \geq 0, \dots, \lambda_k \geq 0$$

18.5 COMPLEMENTARY SLACKNESS CONDITION AND KUHN-TUCKER FORMULATIONS

In this section, we will discuss about the complementary slackness condition and Kuhn Tucker formulations.

18.5.1 Complementary Slackness Condition

Maximize $c^T x$

Subject to $Ax \leq b$

(where T stands for transpose and c, x and b are vectors and A is a matrix)

or,

Minimize $b^T y$ (where y is a vector)

Subject to $A^T y = c$ and $y \geq 0$

Theorem: Let x_0, y_0 be feasible solutions of primal and dual respectively.

Then x_0 and y_0 are optimal if and only if $c^T x_0 = b^T y_0$.

Proof: ($\Rightarrow$) Duality

($\Leftarrow$) Suppose $c^T x_0 = b^T y_0$

(1) Both primal and dual are feasible. Hence both have optima.

(2) $c^T x_0 \leq b^T y_0$ for all feasible solutions of optimal (weak duality)

Since $c^T x_0 = b^T y_0$, x_0 is the optimum of the primal.

(3) Optimum cost of dual = optimum cost of primal.

Again, as $c^T x_0 = b^T y_0$, y_0 is the optimum of dual.

Theorem (Complementary Slackness Condition)